

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

February / March 2024 Semester End Main Examinations

Programme: B.E.

Branch: Computer Science and Engineering

Course Code: 20CS5PCSEG

Course: Software Engineering

Semester: V

Duration: 3 hrs.

Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

UNIT - I

- 1 a) Discuss the responsibilities that the software engineers should have towards professional and society. State ACM/IEEE principles that software engineers should adhere to the code of Ethics and professional practice. **6**
- b) Give reasons for your answers based on the type of system being developed, suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems: **6**
- i) An interactive system that allows flight passengers to find flight times from terminals installed in airports.
 - ii) A banking accounting system that replaces an existing system.
 - iii) A system to control anti-lock braking in a bike.
- c) A software system is to be developed to manage the records of the candidates who register for the Online Course. The record includes the records of all candidates (Course name, duration, start and end dates assignment submission and so on) After the completion of the course the candidate has to take up an exam and qualify the exam with certain minimum percentage for certification. Classify and explain the Non-functional requirements hierarchy diagram for the above software system. **8**

OR

- 2 a) Explain the metrics for specifying non-functional requirements. **6**
- b) Draw the sequence diagram showing the action performed in ATM withdrawal system. You can make any reasonable assumptions about the system. Pay particular attention for specifying user errors. **6**
- c) Illustrate the problems of using natural language for defining user and system requirements, using the example for Insulin Pump Control Software. Explain how structuring natural into forms can help avoid some of these difficulties. **8**

UNIT - II

- 3 a) Identify and analyze the principal viewpoints for the vacation package

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

- reservation service. Organize these using a viewpoint hierarchy diagram with appropriate explanation. **6**
- b) Consider the “Online Fresh Food Store” application where users can place order for the item required using this app. Design the context diagram for this application by showing the various entities that interact with it. Explain your diagram clearly. **6**
- c) Design the state machine model for an automatic washing machine that has different programs for different types of clothes. Give the state and stimulus description for the automatic washing machine. **8**

UNIT - III

- 4 a) Explain the Object Oriented Decomposition for an invoice processing system with a neat diagram. Discuss their advantages and disadvantages. **6**
- b) Using the UML graphical notations identify the object classes, attributes and operations. Also add generalization and aggregation for the following: University Admission Management System. **6**
- c) Analyze the system and suggest an appropriate structural model for the following. Give reasons for your answers. **8**
- An automated ticket-issuing system used by passengers at the railway station.
 - A computer-controlled video conferencing system that allows video, audio and computer data to be visible to several participants at the same time.

Analyze the system and suggest an appropriate control model for the following. Give reasons for your answers

- A television controller that responds to signals from a remote control unit
- A batch processing system that takes information about hours worked and pay rates and also prints salary slips, bank credit transfer information.

UNIT - IV

- 5 a) As a project manager of a company, what are the basic principles you must adopt when you develop a complete schedule for the project? **6**
- b) NETSIM is a network simulation software with an estimated Line of Code (LOC) of 33200. If a team produces an average productivity of 620 LOC per month at the burdened labour rate of \$8000 per month calculate **6**
- Cost per line of code
 - Total cost of the software
 - Estimation in person-months
- c) What types of risks are likely to encounter as the software is being built and as the technical leader how do you assess the overall risk associated with the software? **8**

UNIT - V

- | | | | |
|---|----|---|---|
| 6 | a) | Explain the principles of agile methods. | 6 |
| | b) | Consider that an organization has 20 legacy systems. Analyze how the quality and the business value of each of these systems is assessed and compared with others by plotting it on a chart showing relative business value and system quality. Discuss the different clusters that could be formed. Explain in detail. | 6 |
| | c) | Differentiate between Whitebox and Blackbox testing. | 8 |

OR

- | | | | |
|---|----|---|---|
| 7 | a) | Explain Clean Room Software development with a neat diagram. | 6 |
| | b) | Discuss why program inspections are an effective technique for discovering errors in a program. What types of checks are used to discover errors through inspections? | 6 |
| | c) | Analyze the following Scenario. | 8 |

Payment Collection Process:

Payment may be made in 3 different ways. The user selects which way they wish to pay. If the user has a library subscription, then they can input the subscriber key which should be checked by the system. Alternatively, they can input an organizational account number. If this is valid, a debit of the cost of the article is posted to this account. Finally, they may input the 16-digit credit card number and expiry date. This should be checked for validity and if valid a debit is posted to that credit card account.

Write the Test case description for credit card validity with the following details:

Input, Tests, Output

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

February / March 2024 Semester End Main Examinations

Programme: B.E.

Branch: Computer Science and Engineering

Course Code: 20CS5PCUSP

Course: Unix Shell and System Programming

Semester: V

Duration: 3 hrs.

Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

UNIT - I

- 1 a) With a neat diagram, explain the kernel and shell relationship in UNIX operating System. 5
- b) Consider two files with names flower1 and flower2 in FIRST directory. 5

Flower1	Flower2
Rose	Rose
Daisy	Jasmine
Marigold	Hibiscus
Lily	Lily
Daffodil	Daisy

Write the output of the following command

```
cmp flower1 flower2
comm flower1 flower2
cat flower1 flower2 | wc -l
diff flower1 flower2
cp -R FIRST CLIST
```

- c) i) Identify the syntax error in the script given below. Correct them and find the functionality of the program 10

```
#!/bin/sh
if [ $# -eq 0 ] then
echo "Error - Number missing form command line argument"
echo "Syntax : $0 number"
echo " Use to print multiplication table for given number"
exit 1
fi
n=$1 i=1
while[ $i -le 10]
do
temp=`expr $i \* $n`
echo "$n * $i=temp"
i=`expr $i + 1`
done
```

- ii) Describe about gzip,gunzip,zip and unzip commands with examples to each.

UNIT - II

- 2 a) i) Write a shell script to check whether two arguments passed to scripts are same or not **10**
ii) Write a menu- driven shell script program for implementing a Calculator
- b) i) Write a shell script to print a number in reverse order **10**
ii) Write a shell script to check whether given number is prime or not.

OR

- 3 a) i) Describe the working of set and shift commands in unix, with examples. **10**
ii) Write a shell script to check palindrome of number
- b) i) Describe here document (<<) in unix with examples. **10**
ii) Write a shell script to find factorial of a number.

UNIT - III

- 4 a) i) Explain relative and Absolute permissions of a file with an example to user, group and others. **10**
ii) Differentiate hard links and symbolic links. Illustrate the creation of one link of each type and discuss.
- b) i) Assume stud.lst is a file with details of students in 10 lines. Analyze the functionalities of the commands given below. **10**
- a. touch -m 011522020 stud.lst
 - b. tr -t -d -n stud.lst
 - c. head -n 6 stud.lst | tee shortlist
 - d. cut -d "|" -f 1,4- stud.lst > shortlist.lst
 - e. sort -t "|" -r -k 2 stud.lst

- ii.) Describe about umask and touch commands of unix.

UNIT - IV

- 5 a) Write a C/C++ program to implement copy command of unix using API's. **10**
- b) i) Explain how Unix Kernel supports API's, with a neat diagram **10**
ii) Write a C/C++ POSIX compliant program that prints the POSIX defined Configuration options supported on any given system using feature test macros.

UNIT - V

- 6 a) Describe how processes can be terminated in unix environment, with necessary diagrams. **10**

- b) Write a C/C++ program that demonstrates setjmp and longjmp by creating three user-defined functions say f1, f2, and f3. The function calls are as main() calls f1 which in turn calls f2 which calls f3. Let the user decide if f2 returns back to main or calls f3 which returns back to main. The main function displays which function was called last (f2 or f3) **10**

OR

- 7 a) i) Describe with neat diagram Environment variables. **10**
ii) Describe the different segments of the Memory layout for processes with diagram
- b) Explain the different exec function, and write a C program to perform a word count and printing number of lines using execv function. **10**

B.M.S.C.E. - ODD SEM 2023-24

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

February / March 2024 Semester End Main Examinations

Programme: B.E.

Branch: Computer Science and Engineering

Course Code: 20CS5PEADS

Course: Advanced Data Structures

Semester: V

Duration: 3 hrs.

Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

UNIT - I

- 1 a) Demonstrate the usage of unrolled linked list with an example. **6**
- b) Justify how XOR linked list is better with respect to memory with an example. **6**
- c) Describe how disjoint set can be represented. Analyze the given code to check whether a graph contains a cycle or not using disjoint sets and appropriately complete the code. **8**

```
int find(int parent[], int i)
{
    ----
}
void Union(int parent[], int x, int y)
{
    -----
}
int isCycle( struct Graph* graph )
{
    int *parent = (int*) malloc( graph->V * sizeof(int) );
    memset(parent, -1, sizeof(int) * graph->V);
    for(int I = 0; I < graph->E; ++i)
    {
        int x = find(parent, graph->edge[i].src);
        int y = find(parent, graph->edge[i].dest);
        if (x == y)
            return 1;
        Union(parent, x, y);
    }
    return 0;
}
```

OR

- 2 a) Explain the concept of Smart Union and Path Compression with an example. **6**
- b) Differentiate among the following self-organizing list with an example and **6**

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

justify which method is optimal.

- Count Method
- Transpose Method

- c) Demonstrate and show the contents of the Skip list after inserting the following sequence of elements: 8

- 12 with level 2
- 17 with level 4
- 20 with level 1
- 25 with level 4
- 31 with level 3
- 38 with level 2
- 39 with level 1
- 50 with level 1
- 55 with level 4

After the skip list is constructed, perform the following deletes and update the list. Show steps clearly.

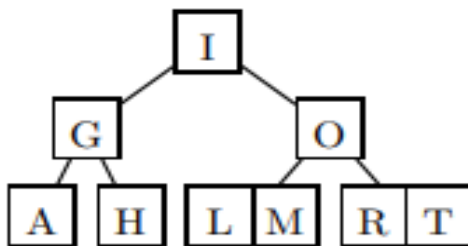
- Delete 17
- Delete 38

UNIT - II

- 3 a) Construct an AVL tree for the following sequence of elements: 8
63, 9, 19, 27, 18, 108, 99, 81

- b) Analyze the below 2-3 tree and show the updated tree contents after the following operations: 6

- Insert 'S'
- Delete 'A'

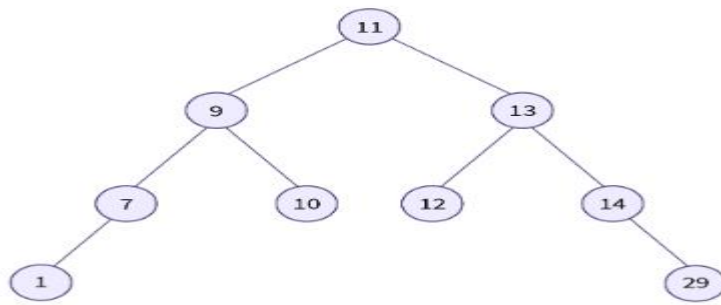


- c) Differentiate between Splay tree and AVL tree. 6

OR

- 4 a) Construct a B-tree of order 5 for the following sequence of characters. After creating the tree, perform Delete 'E' and Delete 'F'. Show the steps clearly. 8
A, F, B, K, D, H, M, J, E, S, I, R, X, C, L, N, T, U, P

- b) Analyze the below Splay tree and identify what type of rotation needs to be taken to splay at element 9. Also, show the updated tree contents after splay at 9. 6



- c) Compare AVL tree and Red Black tree.

6

UNIT - III

- 5 a) Create a Trie data structure for the following strings:

8

“abc”, “abgl”, “cdf”, “abcd”, “lmn”

Also, demonstrate how it searches for the strings: “lmn”, “ghi”

- b) Demonstrate the usage of Fenwick tree with an example

6

- c) Scapegoat tree is a self-balancing binary tree? Justify your answer with an example

6

UNIT - IV

- 6 a) Construct a Hash table for the following sequence of elements:

10

54,26,93,17,77,31,44,55,20

Show how collision is resolved using the following techniques:

- Linear Probing
- Quadratic Probing
- Double Hashing

Consider hash table size $m=11$ and Hash functions:

$H(K)=K\% \text{Table size}$ and $H^1(K)=9 - (K\%9)$

- b) Construct a Hash table for the following numbers:

10

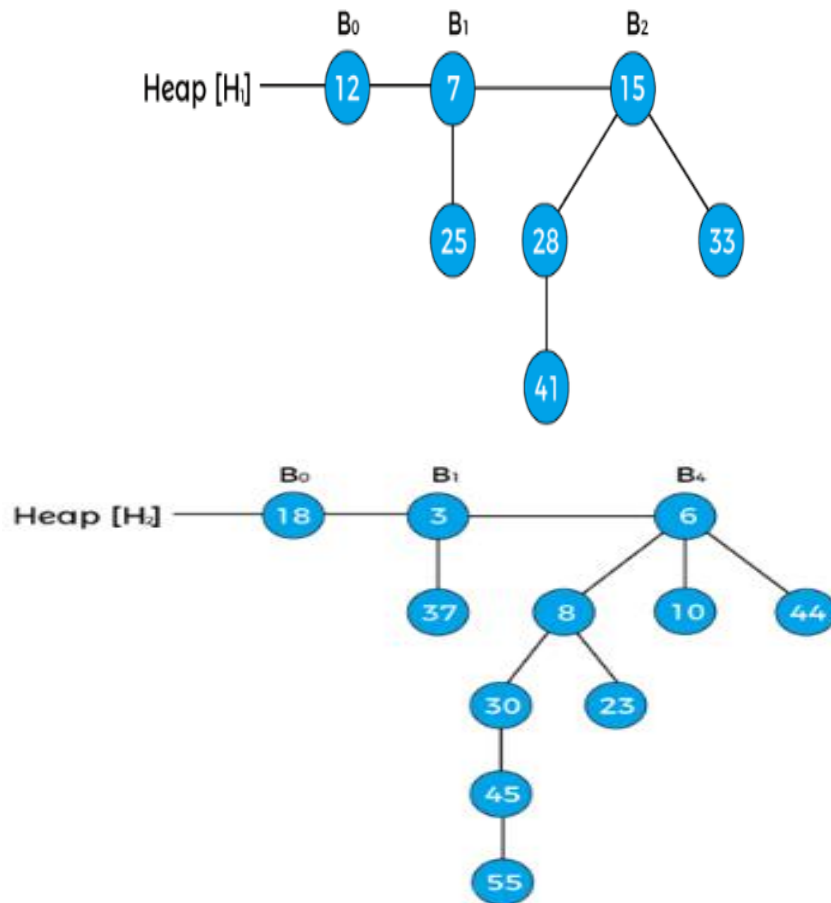
28,4,19,1,22,16,12,0,5,7

Show how collision is resolved using Extendible Hashing with each step demonstrated clearly. Also, mention how directory expansion and bucket splitting taken place. Consider Bucket limit = 3

UNIT - V

- 7 a) Apply union operation on the following two Binomial heaps and show the resultant Binomial heap. Demonstrate each step clearly

10

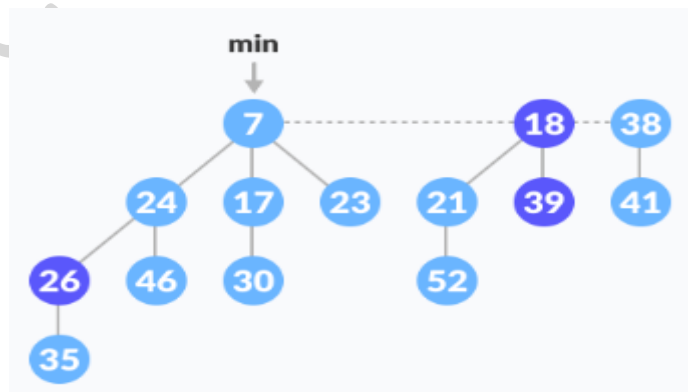


b) For the below Fibonacci Heap, perform the following operations

10

- Decrease the value 46 to 45
- Decrease the value 45 to 15
- Decrease the value 35 to 5
- Delete the value 7

Show all the steps clearly. Note: Marked nodes are 26,18,39



7		A process that models information exchange among several individuals is known as: (a) Initialization (b) Selection (c) Recombination (d) None of the above	CO1	PO1	1
8		The fields of Application of Evolutionary Algorithms are: (a) Combinatorial Optimization (b) Multiple Knapsack Problem (c) Number Partitioning Problem (d) All of the above	CO1	PO1	1
9		Which of the following is not a mutation operator in a genetic algorithm? (a) Random resetting (b) Scramble (c) Inversion (d) Difference	CO1	PO1	1
10		Which of the following statements is/are true? A. Encoding, Fitness function, Selection, CrossOver, Mutation are Genetic Algorithms features. B. Crossover operator defines how chromosomes of parents are mixed in order to obtain genetic codes of their offspring. (a) Statement A is true, Statement B is true (b) Statement A is true, Statement B is false (c) Statement A is false, Statement B is true (d) Statement A is false, Statement B is false	CO1	PO1	1
11		The nucleic acid remaining in the solution can be precipitated by the addition of sodium or ammonium acetate and ethanol. (a) True (b) False	CO1	PO1	1
12		Fitness function should be (a) maximum (b) minimum (c) intermediate (d) none of these	CO1	PO1	1
13		Genetic algorithms are example of (a) heuristic (b) Evolutionary algorithm (c) ACO(d) PSO	CO1	PO1	
14		Which of the following statement is true in terms of partial digestion? (a) Partial digestion can be defined as a condition which does not recognize any of the site present in the DNA sequence. (b) Partial digestion created a similar number of fragments to that of complete digestion. (c) Partial digestion does not help represent a genomic library (d) It can identify exactly half of the sites in the DNA	CO1	PO1	1

15		Which of the following activity can't be seen in the case of DNA polymerase? (a) 3'-5' exonuclease (b) 5'-3' exonuclease (c) 5'-3' DNA synthesis (d) 3'-5' DNA synthesis	CO1	PO1	1
16		_____ enzyme helps in carrying out elements of mobile genetics from one portion of DNA to another. (a) Transposase (b) Endonuclease (c) Ligase (d) Transcriptase	CO1	PO1	1
17		Pheromone quantity in ACO is ____ proportional to path selection. (a) directly (b) inversely (c) not (d) none of the above	CO1	PO1	1
18		The algorithm in which the individuals evaluate themselves, compare to their neighbors and imitate only those neighbors who are superior to themselves: (a) Particle Swarm (b) Ant Colony (c) Immune Computing (d) Artificial Neural Networks	CO1	PO1	1
19		Arrange the following steps in genetic algorithm in order: 1. random selection of initial population 2. crossover and mutation 3. encoding of chromosome 4. selection of best parental chromosome (a) 4, 1, 2, 3 (b) 3, 1, 4, 2 (c) 3, 1, 2, 4 (d) 1, 3, 2, 4	CO1	PO1	1
20		Swarm intelligence is a type of: (a) Computing with Natural Materials (b) Simulation and Emulation of Nature in Computers (c) Computing Inspired by Nature (d) Calculus Based Method	CO1	PO1	1
PART B					
1	a)	Describe the different types of chromosomes. Write the functions of a nucleus.	CO1	PO1	5

	b)	Write the Mendel's Laws of Genetics with Mendelian Crosses as an example.	CO1	PO1	5
2	a)	With a neat diagram, discuss the molecular structure of a DNA.	CO1	PO1	5
	b)	Why Evolutionary computation is important? Explain the working of Evolutionary Computation.	CO1	PO1	5
3	a)	Explain the process of Recombination and Mutation in dissecting an evolutionary algorithm.	CO1	PO1	5
	b)	Describe the transcription and translation process in Gene Expression.	CO1	PO1	5
4	a)	Explain Ant Colony Optimization algorithm with an example.	CO1	PO1	5
	b)	What is RNA Splicing? Explain the process of Spliceosome.	CO1	PO1	5
5	a)	Explain the working of Particle Swarm Optimization.	CO1	PO1	5
	b)	Discuss the various types of NanoMolecules in Biomedical Sciences.	CO1	PO1	5

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

February / March 2024 Semester End Main Examinations

Programme: B.E.

Branch: Computer Science and Engineering

Course Code: 22CS5HSSPM

Course: Software Project Management and Finance

Semester: V

Duration: 3 hrs.

Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

UNIT - I

- 1 a) Define Project. Explain the characteristics of Project. **6**
- b) Campbell Industries has a project with the following projected cash flows: **8**
 Initial Cost, Year 0: \$468,000
 Cash flow year one: \$135,000
 Cash flow year two: \$240,000
 Cash flow year three: \$185,000
 Cash flow year four: \$135,000
 Should this project be accepted or rejected for the following discount rates?:
 i) 8%
 ii) 14%
 iii) 20%
 Use NPV model for all the above discount rates.
- c) The management of Fine Electronics Company is planning to purchase an equipment to be attached with the main manufacturing machine. The equipment will cost \$6,000 and will increase annual cash inflow by \$2,200. The useful life of the equipment is 6 years. The management wants a 20% return on all investments. **6**
 i) Compute net present value (NPV) of this investment project.
 ii) Should the equipment be purchased according to NPV analysis?

OR

- 2 a) List and explain the skills the Project Manager needs to have to maintain a good environment in a Project and also list the roles of a Project Manager. **6**
- b) What are the Payback Periods of Projects E and F? Assume all cash flows are evenly spread throughout the year. If the cut-off period is three years, which projects do you accept? **8**

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

<i>Projects</i>	<i>E</i>	<i>F</i>
<i>Cost</i>	\$40,000	\$250,000
<i>Cash Flow Year One</i>	\$10,000	\$40,000
<i>Cash Flow Year Two</i>	\$10,000	\$120,000
<i>Cash Flow Year Three</i>	\$10,000	\$200,000
<i>Cash Flow Year Four</i>	\$10,000	\$200,000
<i>Cash Flow year Five</i>	\$10,000	\$200,000
<i>Cash Flow Year Six</i>	\$10,000	\$200,000

- c) A company wants to establish its branch in Jammu and Kashmir. Analyze how the environment affects the company and discuss on the same. **6**

UNIT - II

- 3 a) List and explain the components of Project Management Plan. **6**
b) Write the Risk plan Script for “Online Food ordering System”. **7**
c) Write the Scope and Vision Statement for Library Management System. **7**

UNIT - III

- 4 a) Define Statement of work. List and explain the contents of SOW. **6**
b) Design a Charter for developing a website for Online course registration System. **8**
c) Discuss how the quality of the Project is related to triple Constraints. Construct the Priority Matrix for the scheduled maintenance of a banking website and analyze how the triple constraints are affected with respect to the given system. Justify your answer. **6**

UNIT - IV

- 5 a) List and explain the necessary features for cost estimation formula with the steps of Cost Estimation process, and explain the types of cost estimates. **6**
b) A product based company has estimated what they need the following staff: 4 investors where each of them will make 7,00,000 INR per year and 5 technical staffs where each of them will make 3,00,000 INR per year. All the Employees will get company benefits with 1 week of vacation, 10 Holidays and 10 sick leaves, pension plan that costs the company 5% of the labour cost and medical plan that costs 30% of the labour cost. **10**

The investors estimate that they need to borrow a loan of rupees 30,00,000 INR with 8% interest and they plan to pay off the loan in 6 years. They require 15,000 square feet of space, rented at 1000 INR per square foot per year. Software and Hardware equipments are estimated at 50% of the space rental. Payroll, legal and other corporate expenses are estimated at 5% per year. One of the partners performed a study and believes their product will be competitive at 30,000 INR per product and they will be able to sell 1,000 products.

The product costed 10,00,00,000 INR each. The partners plan for a profit percentage of 50%. Calculate the Total Direct labour, Fringe benefits, Total Overhead cost, Gross Profit, Net profit. Is this a profitable venture as per their survey?

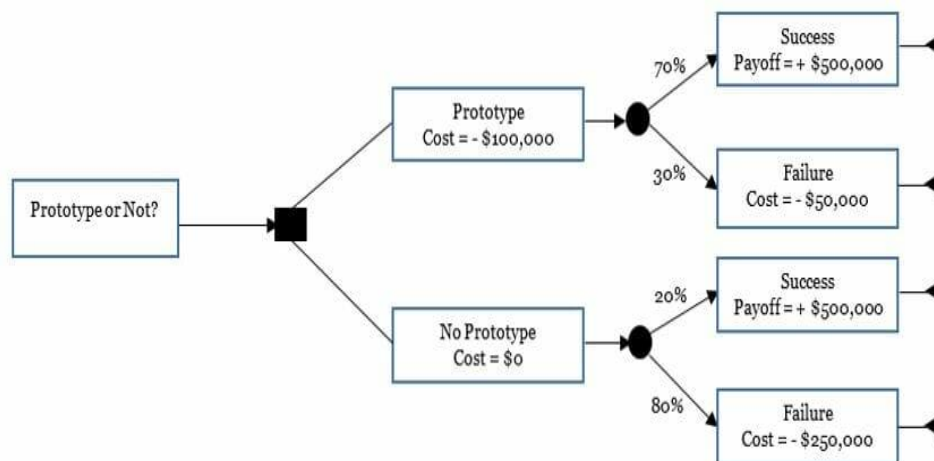
- c) Differentiate between Delphi Technique vs. Analogous Estimation Technique. 4

OR

- 6 a) Define Parametric Estimate. Explain the classification of data Elements. 6
- b) i) You are implementing a website for online shopping. After analyzing the scope, you decide that you need 2 login forms for customer name and email-id, 2 forms for the customer address and other details. We need one form to print out the details of customer. As to data entities, we need 20 for customer name and email-id, 20 for their address and 20 with each purchase. Calculate the cost estimate for the Programming effort and Total Effort. 10
- ii) The team estimates the time to develop a new system as follows: Most likely, the project takes 4 months. In the worst case, the completion takes 7 months, if there are no hurdles, the completion takes 2 months. Calculate the PERT Mean and Project Schedule Estimate and Discuss about the Confidence Level.
- c) Analyze the efficiency of Bottom-up cost estimation technique in comparison with Top-down cost estimation with an example. 4

UNIT - V

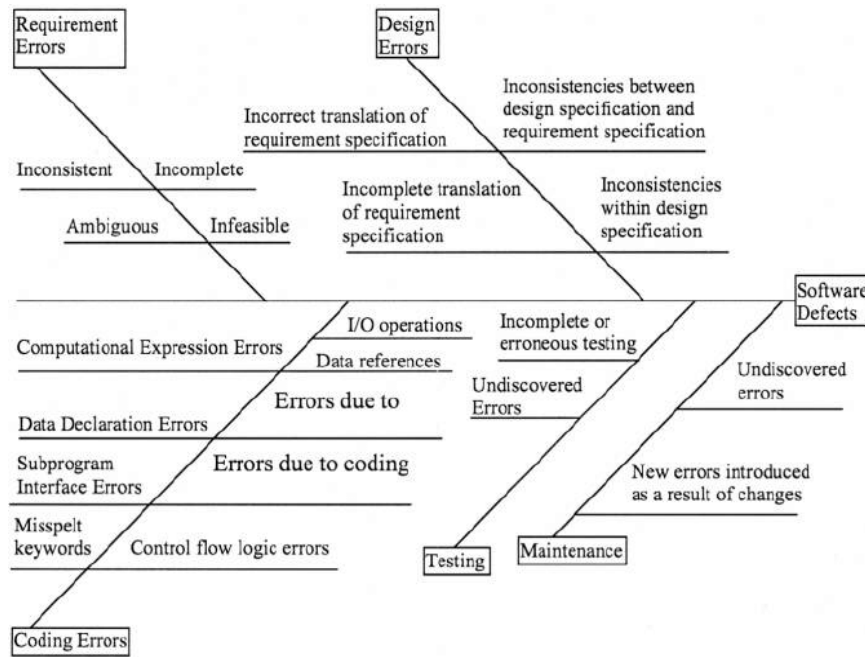
- 7 a) List and explain the Risk Strategies for Negative and Positive Risks. 6
- b) 7



Analyze the above Decision tree and Explain the same with the notations used in the diagram. Calculate the EMV value of each branch of the tree. Which one would you select? Justify the reason for the same.

c)

7



Analyze how the above fishbone diagram helps in identifying the cause and effect for the problem and also explain the different notations used in the above diagram.

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B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

February / March 2024 Semester End Main Examinations

Programme: B.E.

Branch: Computer Science and Engineering

Course Code: 22CS5PCAIN / 20CS5PCAIP

Course: Artificial Intelligence

Semester: V

Duration: 3 hrs.

Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.			UNIT - I	CO	PO	Marks
	1	a)	Implement a Utility based agent for the vacuum environment. Assume that initially the environment with this agent has all possible square as dirty.	CO1	PO1	05
		b)	The missionaries and cannibal's problem are usually stated as follows. Three missionaries and three cannibals are on one side of a river, along with a boat that can hold one or two people. Find a way to get everyone to the other side without ever leaving a group of missionaries in one place outnumbered by the cannibals in that place. i. Formulate the problem precisely, making only those distinctions necessary to ensure a valid solution. ii. Why do you think people have a hard time solving this puzzle, given that the state space is so simple?	CO1	PO1	10
		c)	Analyze the PEAS Description for: i. Part Picking Robot ii. Satellite Image analysis System	CO2	PO2	05
			OR			
	2	a)	You are provided with three jugs: one having the capacity to hold 3 gallons of water, 5 gallons of water and the other has the capacity to hold 8 gallons of water. There is no other measuring equipment available and the jugs also do not have any kind of marking on them. So, the agent's task here is to fill the 8-gallon jug with 4 gallons of water (Jug A, Jug B) by using only these three jugs and no other material. Initially, both our jugs are empty. Explain this Water Jug Problem using state space search. Generate Production rules for this problem.	CO2	PO2	05

		8 liter (Jug A) 5 liter (Jug B) 3 lite (Jug C)																					
	b)	Analyze the PEAS Description for: i. Ketchup producing Industry iv. Medical Diagnosis system	CO2	PO2	10																		
	c)	Infer a suitable agent architecture for an automated taxi driver. Justify your answer.	CO1	PO1	05																		
		UNIT - II																					
3	a)	Apply informed heuristic A* search algorithm for the given problem. Whether it is complete and optimal? What is the time and space complexity? Each edge has an associated weight & heuristic cost is given in parenthesis inside each node.	CO1	PO1	10																		
	b)	Given the 8-puzzle shown in Figure below, use the hill-climbing algorithm with the Manhattan distance heuristic to find a path to the goal state. <table><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>8</td><td></td></tr><tr><td>7</td><td>6</td><td>5</td></tr></table> Initial state <table><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td><td>6</td></tr><tr><td>7</td><td>8</td><td></td></tr></table> Goal state	1	2	3	4	8		7	6	5	1	2	3	4	5	6	7	8		CO1	PO1	10
1	2	3																					
4	8																						
7	6	5																					
1	2	3																					
4	5	6																					
7	8																						

		UNIT - III																																								
4	a)	Illustrate the resolution algorithm step by step and Prove that the sentence S is true in the given Propositional logic sentence: $(P \wedge Q) \wedge (P \Rightarrow R) \wedge [(Q \wedge R) \Rightarrow S]$	CO3	PO3	10																																					
	b)	Represent the following statements into First order logic representation. i) All parrots are birds and all birds are animals. ii) All plants are green or brown. iii) The widow slaps every woman who does not slap herself. iv)Some students are not smart. v) No birds are reptiles.	CO2	PO2	10																																					
		OR																																								
5	a)	Consider the following Wumpus world problem and prove that the Wumpus is in the room (1, 3). Consider the knowledge base given for the reference. Representation of Knowledge Base for Wumpus world is as given below: <table><tr><td>$\neg W_{11}$</td><td>$\neg S_{11}$</td><td>$\neg P_{11}$</td><td>$\neg B_{11}$</td><td>$\neg G_{11}$</td><td>V_{11}</td><td>OK_{11}</td></tr><tr><td>$\neg W_{12}$</td><td>----</td><td>$\neg P_{12}$</td><td>-----</td><td>----</td><td>$\neg V_{12}$</td><td>OK_{12}</td></tr><tr><td>$\neg W_{21}$</td><td>$\neg S_{21}$</td><td>$\neg P_{21}$</td><td>B_{21}</td><td>$\neg G_{21}$</td><td>V_{21}</td><td>OK_{21}</td></tr></table> Following is the Simple KB for Wumpus world when an agent moves from room [1, 1], to room [2,1], <table><tr><td>1,4</td><td>2,4 P?</td><td>3,4</td><td>4,4</td></tr><tr><td>1,3 W?</td><td>2,3 S G B</td><td>3,3</td><td>4,3</td></tr><tr><td>1,2</td><td>2,2 V P?</td><td>3,2</td><td>4,2</td></tr><tr><td>1,1 A ok</td><td>2,1 B V ok</td><td>3,1 P?</td><td>4,1</td></tr></table>	$\neg W_{11}$	$\neg S_{11}$	$\neg P_{11}$	$\neg B_{11}$	$\neg G_{11}$	V_{11}	OK_{11}	$\neg W_{12}$	----	$\neg P_{12}$	-----	----	$\neg V_{12}$	OK_{12}	$\neg W_{21}$	$\neg S_{21}$	$\neg P_{21}$	B_{21}	$\neg G_{21}$	V_{21}	OK_{21}	1,4	2,4 P?	3,4	4,4	1,3 W?	2,3 S G B	3,3	4,3	1,2	2,2 V P?	3,2	4,2	1,1 A ok	2,1 B V ok	3,1 P?	4,1	CO3	PO3	10
$\neg W_{11}$	$\neg S_{11}$	$\neg P_{11}$	$\neg B_{11}$	$\neg G_{11}$	V_{11}	OK_{11}																																				
$\neg W_{12}$	----	$\neg P_{12}$	-----	----	$\neg V_{12}$	OK_{12}																																				
$\neg W_{21}$	$\neg S_{21}$	$\neg P_{21}$	B_{21}	$\neg G_{21}$	V_{21}	OK_{21}																																				
1,4	2,4 P?	3,4	4,4																																							
1,3 W?	2,3 S G B	3,3	4,3																																							
1,2	2,2 V P?	3,2	4,2																																							
1,1 A ok	2,1 B V ok	3,1 P?	4,1																																							
	b)	Convert the following into FOL i) There are no leafy vegetables that are tasty and starchy. ii) Anyone who kills an animal is loved by no one. iii) Either everything is bitter or everything is sweet. iv) All animals either eat all plants or eat all smaller animals that eat some plants. v) All the courses in the Food Processing department are easy.	CO2	PO2	10																																					

			UNIT - IV			
6	a)	Conversion of Facts into FOL Prove by resolution that: “John likes peanuts” a. John likes all kind of food. b. Apple and vegetable are food c. Anything anyone eats and not killed is food. d. Anil eats peanuts and still alive e. Harry eats everything that Anil eats.	CO3	PO3	10	
	b)	Infer whether the following pairs statements can be unified or not. If unification is possible, write the substitutions. Justify your answer. i) Human (Marcus) and Human (Julius) ii) Student(x) and Teacher(y) iii) Knows (John, F(x)) and Knows (y, F(G(y)) iv) Likes (Ram, x) and Likes (x, Raj) v) Loves(x,y) and Loves (z)	CO2	PO2	05	
	c)	The Knowledge Base contains: – allergies(X) $\rightarrow$ sneeze(X) – cat(Y) $\wedge$ allergicToCats(X) $\rightarrow$ allergies(X) – cat(felix) – allergicToCats(mary) Resolve by resolution for the goal state sneeze (mary)	CO3	PO3	05	
		UNIT - V				
7	a)	Consider a bag of three biased coins a, b, and c with probabilities of coming up heads of 20%, 60%, and 80%, respectively. One coin is drawn randomly from the bag (with equal likelihood of drawing each of the three coins), and then the coin is flipped three times to generate the outcomes X1, X2, and X3. a. Draw the Bayesian network corresponding to this setup and define the necessary Conditional Probability tables (CPTs). b. Calculate which coin was most likely to have been drawn from the bag if the observed flips come out heads twice and tails once.	CO3	PO3	10	
	b)	Write the variable elimination algorithm and show how the optimization is achieved with respect to computations done.	CO1	PO1	10	

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B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

February / March 2024 Semester End Main Examinations**Programme: B.E.****Branch: Computer Science and Engineering****Course Code: 22CS5PCCPD****Course: Compiler Design****Semester: V****Duration: 3 hrs.****Max Marks: 100**

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.			UNIT - I	CO	PO	Marks
	1	a)	Analyze the following line of code and show how this line is converted to target code in assembly language over the various phases of compiler. $P = qr * rs + 70.00$	CO2	PO2	7
		b)	Construct a transition diagram to represent relational operators. Give a code segment for the implementation of a relational operator.	CO1	PO1	10
		c)	Differentiate between Interpreter and Compiler.	CO2	PO2	3
			UNIT - II			
	2	a)	Eliminate Left recursion from the following grammar. $E \rightarrow E + T \mid E - T \mid T$ $T \rightarrow T * F \mid T / F \mid F$ $F \rightarrow (E) \mid id$	CO1	PO1	5
		b)	Analyze and design a Predictive Parser for the grammar. $S \rightarrow (L) \mid a$ $L \rightarrow L, S \mid S$ Also Construct the Predictive Parsing table.	CO2	PO2	5
		c)	Design a LR(0) Parser for the grammar $S \rightarrow aSbS$ $S \rightarrow bSaS$ $S \rightarrow \epsilon$ And also parse the string "aabbab" give transition diagram.	CO2	PO2	10
			OR			
	3	a)	Given the grammar $A \rightarrow (A)$ $A \rightarrow a$ i. Find LR(0) items ii. Give transition diagram	CO2	PO2	10

		iii. Construct SLR(1) Parsing table. iv. Show the Parsing steps for the string “((a))”			
	b)	Design LR(1) Parser and LALR parser for the grammar. $S \rightarrow 0S1$ $S \rightarrow 01$ and also parse the string “000111” for LALR parser.	CO2	PO2	10
		UNIT - III			
4	a)	Consider the Context free grammar given below $S \rightarrow En$ $E \rightarrow E+T \mid E-T \mid T$ $T \rightarrow T*F \mid T/F \mid F$ $F \rightarrow (E) \mid \text{digit}$ $n \rightarrow ;$ where n represents end of file marker i) Obtain the SDD for the above grammar. ii) Construct the parse tree, syntax tree and annotated parse tree for the input string $5*6+7$;	CO1	PO1	8
	b)	Give the Syntax directed translation of type $\text{int}[2][3]$ and also given the Semantic rules for the respective productions.	CO1	PO1	8
	c)	List the differences between Synthesized and Inherited attributes.	CO2	PO2	4
		UNIT - IV			
5	a)	Explain Static Single Assignment form(SSA).With an Example.	CO1	PO1	6
	b)	Give the three address code for the following i. $a = b * - c + b * - c$ ii. $\text{if}(A < B) \text{ then } 1$ $\text{else } 0$	CO3	PO3	6
	c)	Translate $a = b * - c + b * - c$ into Quadruple , triple and indirect triple.	CO3	PO3	8
		OR			
6	a)	Give Syntax directed translation for switch statement.	CO2	PO2	6
	b)	Obtain DAG for $a + a * (b-c) + (b-c) * d$.Also give steps for Construction of DAG.	CO3	PO3	8
	c)	Write the three address code for the following: In both position number and Symbolic labels. <pre> C=0 do{ if(a<b) then x++; else {x - -;} c + + }while(c<5) </pre>	CO3	PO3	6

			UNIT - V			
7	a)	Generate simple machine code for the following three address code. Show the step by step contents of address descriptor and register descriptor. Assume there are three registers R1, R2, R3. Let a,b,c,d be live on exit from the block. Let t,u,v be temporary to the block. t=a-b u=a-c v=t+u a=d d=v+u	CO2	PO2	10	
	b)	Generate three address code , Basic Block and flow graph for i=1 do Sum = sum + a[i] * b[i] While(i<=20)	CO3	PO3	10	

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B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

February / March 2024 Semester End Main Examinations

Programme: B.E.

Branch: Computer Science and Engineering

Course Code: 22CS5PCCRP / 20CS6PCCNS

Course: Cryptography / Cryptography and Network Security

Semester: V

Duration: 3 hrs.

Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.			UNIT - I	CO	PO	Marks
	1	a)	Demonstrate with a suitable example how mono alphabetic substitution cipher is vulnerable to frequency analysis attack.	CO1	PO1	5
		b)	Given that $Z_7 = \{1, 2, 3, 4, 5, 6\} * \text{mod } 7$ is a group, write all the cyclic subgroups of different orders of Z_7 . Is Z_7 a cyclic group?	CO1	PO1	10
		c)	The ciphertext GEZXDS was encrypted by a Hill cipher with a $2 * 2$ matrix. The plaintext is "solved". Find the encryption matrix M.	CO1	PO1	5
			OR			
	2	a)	Encrypt the message "the house is being sold tonight" using the following ciphers. Ignore the space between words. Decrypt the message to get the plaintext: a. Vigenere cipher with key: "dollars" b. Autokey cipher with key = 7 c. Affine Cipher with key (15,20)	CO1	PO1	10
		b)	Show the steps involved in multiplication of two polynomials: $f(x) = x^6 + x^4 + x^2 + x + 1$ and $g(x) = x^7 + x + 1$ in $GF(2^8)$ using the efficient algorithm for multiplication using n-bit words. Consider $x^8 + x^4 + x^3 + x + 1$ as the irreducible polynomial.	CO1	PO1	10
			UNIT - II			
	3	a)	Draw and explain Mix Columns transformation in AES with an example	CO1	PO1	5
		b)	Explain key Generation in DES with a neat figure.	CO1	PO1	10
		c)	Illustrate how double DES is vulnerable to meet-in-the-middle attack.	CO2	PO2	5
			UNIT - III			
	4	a)	Apply CRT to find the integer x which leave a remainder of 6, 13, 9 and 19 when divided by 11, 16, 21 and 25 respectively.	CO1	PO1	10

	b)	Using quadratic residues, solve the following congruences: a) $x^2 \equiv 4 \pmod{7}$ b) $x^2 \equiv 5 \pmod{11}$ c) $x^2 \equiv 7 \pmod{13}$ d) $x^2 \equiv 12 \pmod{17}$	CO1	PO1	10
		UNIT - IV			
5	a)	Explain with an example the working of SHA algorithm.	CO1	PO1	8
	b)	In ElGamal cryptosystem, given the prime $p = 31$: a. Choose an appropriate values for e_1 and d , then calculate e_2 . b. Encrypt the message "HELLO". Use 00 to 25 for encoding. Use different blocks to make $P < p$. c. Decrypt the ciphertext to obtain the plaintext. Clearly show all the steps involved in encryption and decryption.	CO1	PO1	12
		OR			
6	a)	Encrypt the message block $M = 2$ using RSA with the following parameters: $e = 23$ and $n = 233 \times 241$. Compute a private key (d, p, q) corresponding to the given above public key (e, n)	CO3	PO3	5
	b)	Consider the elliptic curve $E_{11}(1,6)$; that is, the curve is defined by $y^2 = x^3 + x + 6$ with a modulus of $p=11$. i. Determine all the points in $E_{11}(1,6)$. ii. Consider the point $G = (2,7)$. Compute the multiple of G from $2G$ through $4G$	CO3	PO3	10
	c)	Demonstrate cycling attack for the RSA cryptosystem.	CO1	PO1	5
		UNIT - V			
7	a)	Demonstrate with an example how digital signature satisfies the following properties: (1) Message authentication (2) Message integrity (3) Nonrepudiation (4) Confidentiality	CO2	PO2	8
	b)	Using the RSA Digital Signature scheme, let $p = 809$, $q = 751$ and $d = 23$. Calculate the public key e . Then do the following: a. Sign and verify a message with $M_1 = 101$ Calculate the signature S_1 . b. Sign and verify a message with $M_2 = 51$. Calculate the signature S_2 . c. Show that if $M = M_1 \times M_2 = 5151$, then $S = S_1 \times S_2$.	CO1	PO1	12

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B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

February / March 2024 Semester End Main Examinations

Programme: B.E.

Branch: Computer Science and Engineering

Course Code: 22CS5PCIOT / 20CS5PEIOT

Course: Internet of Things

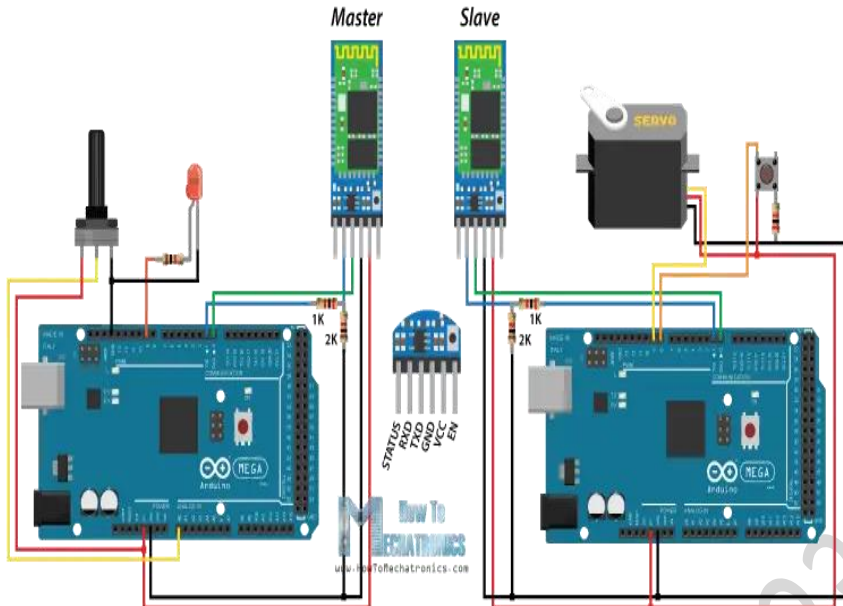
Semester: V

Duration: 3 hrs.

Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.			UNIT - I	CO	PO	Marks
	1	a)	Describe REST based communication API and its Architectural Constraints	CO1	PO1	06
		b)	Apply the knowledge of different IOT levels and Analyze the suitable IoT levels for designing Smart irrigation system and package tracking system	CO2	PO2	08
		c)	Design and implement an IOT system for smart home where the lights are ON/OFF based on light intensity and presence of human in the room.	CO3	PO3	06
			OR			
	2	a)	Illustrate the features and pin configuration of Arduino Board	CO3	PO3	06
		b)	Design and Implement a smart irrigation system where depending on the moisture levels a motor is switched ON/OFF and if a animal enters the farm an alarm is raised to the owner of the farm	CO3	PO3	08
		c)	Identify and list the sensors used to sense closeness of an object relative to another object and illustrate its working principle	CO3	PO3	06
			UNIT - II			
	3	a)	Write a neat block diagram of IOT Reference Architecture Model and explain the functional responsibility of layer that convert network data flows into information that is suitable for storage and higher level processing	CO2	PO2	06
		b)	Analyze the circuit given below, List the components and Write the server side and client-side program to make the circuit operational.	CO3	PO3	08

					
	c)	Write and explain the commands to configure two ESP8266 modules as access point and station respectively. Use this Configuration to switch on the buzzer from the commands received from the station	CO3	PO3	06
		UNIT - III			
4	a)	Describe the architecture of Software defined networks and the functionality of each layer	CO2	PO2	06
	b)	Design and implement a Raspberrypi program to control the display of LED through push button.	CO3	PO3	08
	c)	Demonstrate how CoAP protocol is different from HTTP protocol and illustrate the method to achieve reliability in CoAP	CO1	PO1	06
		OR			
5	a)	Explain features of 6LoWPAN adaptation layer and mesh addressing headers in the context of 802.15.4 network	CO1	PO1	06
	b)	Analyze the Type of messages exchanged in RPL. Construct a RPL DODAG for the topology with OF is to minimize the ETx and explain the steps.	CO1	PO1	08

	c)	Assume that a Garage door opener system has a light controller which can control the locking of the Garage door Write a sequence to register the light controller in a server. Support your design with appropriate sequence diagram.	CO1	PO1	06
		UNIT - IV			
6	a)	Identify and explain the IOT cloud services with examples	CO1	PO1	06
	b)	Demonstrate Edge analytics with the framework for analytics in IOT and explain the two types of IOT Analytics	CO2	PO2	08
	c)	Differentiate between Database Management system and Data stream management system at Data level, processing level and query level.	CO2	PO2	06
		UNIT - V			
7	a)	Paraphrase the importance of Data analytics in IOT.	CO2	PO2	06
	b)	Describe Feature selection using wrapper based technique and write a python program considering 5 parameters of air pollution dataset	CO2	PO2	08
	c)	Identify and explain the key challenges of data stream mining as compared to traditional data mining.	CO2	PO2	06

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Autonomous Institute Affiliated to VTU

February / March 2024 Semester End Main Examinations

Programme: B.E.

Branch: Computer Science and Engineering

Course Code: 22CS5PEAAM / 20CS5PEAAG

Course: Advanced Algorithms

Semester: V

Duration: 3 hrs.

Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

		UNIT - I	CO	PO	Marks																						
1	a)	Apply dynamic programming technique to find edit distance between S= “hello” and T= “keep”.	CO1	PO1	5																						
	b)	Justify how Matrix Chain Multiplication reduces number of multiplications with an example.	CO2	PO2	5																						
	c)	Design dynamic programming based algorithm for solving rod cut problem. Apply the same for the below instance and find solution for rod of length=8. <table><tr><td>Length i</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Price Pi</td><td>1</td><td>5</td><td>8</td><td>9</td><td>10</td><td>17</td><td>17</td><td>20</td><td>24</td><td>30</td></tr></table>	Length i	1	2	3	4	5	6	7	8	9	10	Price Pi	1	5	8	9	10	17	17	20	24	30	CO1,3	PO1,3	10
Length i	1	2	3	4	5	6	7	8	9	10																	
Price Pi	1	5	8	9	10	17	17	20	24	30																	
		UNIT - II																									
2	a)	Apply Ford-Fulkerson algorithm to find maximum flow in the below network. Also write Ford-Fulkerson algorithm. <pre>graph LR; s((s)) -- 16 --> v1((v1)); s -- 13 --> v2((v2)); v1 -- 12 --> v3((v3)); v2 -- 10 --> v1; v2 -- 14 --> v4((v4)); v3 -- 4 --> v2; v3 -- 20 --> t((t)); v4 -- 9 --> v3; v4 -- 4 --> t</pre>	CO1	PO1	8																						
	b)	Design a merge sort algorithm to sort n numbers for: i. Merge is not multithreaded ii. Merge is also multithreaded compare speedup achieved in both cases.	CO2,3	PO2,3	12																						
		OR																									

3	a)	Design an algorithm for Multithreaded Matrix Multiplication. What is the speedup achieved with this?	CO2,3	PO2,3	10
	b)	How Ford-Fulkerson algorithm can be used to solve Maximum Bipartite problem? Explain with an example.	CO2	PO2	6
	c)	Explain the use of keywords “spawn” and “sync” in Multithreaded programming with an example.	CO1	PO1	4
		UNIT - III			
4	a)	Apply Rabin Karp algorithm to find Pattern P= “215” in Text T= “33456732156”. Use Mod 11 and also find number of spurious hits. Also compare Naïve string matching with Rabin Karp algorithm.	CO1,2	PO1,2	10
	b)	Design an algorithm based on Finite Automata for searching a pattern P in a given text T. Apply the same for Pattern P= “abaab” and Text T= “ababbaaabaabab”.	CO1,3	PO1,3	10
		OR			
5	a)	Write KMP string matching algorithm. Apply the same for finding Pattern P= “abaab” in Text T= “ababbaaabaab”.	CO1,3	PO1,3	10
	b)	Design pseudo code for Horspool string matching. Apply the same for Text T= “JIM_SAW_ME_IN_BARBERSHOP” and Pattern P= “BARBER”.	CO1,3	PO1,3	10
		UNIT - IV			
6	a)	Convert the below LPP to Standard form: Minimize 2a-b Subject to a - b ≤ 0 a + b > 0 a ≥ 0	CO1	PO1	5
	b)	Formulate Maximum Flow problem as LPP.	CO1	PO1	5
	c)	Solve the below LPP using Simplex method: Maximize $z=2x_1-x_2+2x_3$ subject to $2x_1+x_2 \leq 10$ $x_1+2x_2-2x_3 \leq 20$ $x_2+2x_3 \leq 5$ $x_1, x_2, x_3 \geq 0$	CO1	PO1	10
		UNIT - V			
7	a)	Design pseudo code for checking whether a pair of line segments intersect or not. Apply the same to check line segment (P1,P2) intersects with (P3,P4). P1= (10,10) P2=(10,30), P3=(30,30) and P4=(40,40).	CO1,3	PO1,3	10
	b)	Explain the working of Jarvis’s March algorithm for finding Convex Hull with an example.	CO1	PO1	10

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February / March 2024 Semester End Main Examinations**Programme: B.E.****Branch: Computer Science and Engineering****Course Code: 22CS5PEDEV****Course: Data Exploration and Visualization****Semester: V****Duration: 3 hrs.****Max Marks: 100**

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.			UNIT - I	CO	PO	Marks
	1	a)	Discuss the common data types used to represent columns in a dataset with an example. What are the different measurement scales for these data types and possible statistics can be performed on it?	CO1	PO1	10
		b)	Discuss the steps in Exploratory Data Analysis.	CO1	PO1	10
			UNIT - II			
	2	a)	You are given a dataset containing information about customers' orders. The dataset has the following columns. The dataset may contain missing values and duplicates. Order_ID: Unique identifier for each order. Customer_ID: Unique identifier for each customer. Product_ID: Identifier for the purchased product. Quantity: Number of units of the product ordered. Order_Date: Date when the order was placed. Total_Amount: Total amount spent on the order. - Find duplicate rows in the 'Order_ID' column of the DataFrame and remove the duplicates. - To find the total number of missing values in the dataset - Drop rows which contains all null values. - Drop rows where Quantity columns have missing values. - Check for missing values in a Keep the row which has Threshold (3) of NoT NULL values.	CO2	PO2	10
		b)	A dataset containing information about the temperature readings for different days. The dataset has the following columns: Date: The date of the temperature reading.	CO2	PO2	05

		'Date': ['2021-01-01', '2021-01-02', '2021-01-03', '2021-01-04', '2021-01-05', '2021-01-06'], Temperature: The recorded temperature for each day. 'Temperature': [25, 28, np.nan, 22, np.nan, 30] Perform the following • Load the dataset into a DataFrame. • Explore the pattern of temperature fluctuations and based on the pattern observed in the temperature choose whether forward filling or backward filling is more suitable for this dataset. • Provide a brief justification for your choice.			
	c)	Suppose you have a dataset containing information about customers' purchases at a store. The dataset (customer_data.csv) includes columns: 'Customer_ID', 'Age', 'Gender', 'Purchase_Amount'. Your task is to perform random sampling to select a subset of 20 customers from this dataset for a survey.	CO1	PO1	05
		UNIT - III			
3	a)	You are given a dataset containing information about sales transactions. The dataset has the following columns: Product: The name of the product. Category: The category to which the product belongs. Price: The price of the product. Quantity: The quantity of the product sold. perform the following tasks using the groupby() function: • Calculate the total revenue (Price * Quantity) for each product. • Find the average price for each category. • Determine the product with the highest total revenue.	CO1	PO1	10
	b)	Define the terms "skewness" and "kurtosis." How do they describe the shape of a distribution?	CO1	PO1	10
		UNIT - IV			
4	a)	Explain the characteristics of time series data.	CO1	PO1	06
	b)	Discuss the statement "Correlation does not imply causation".	CO1	PO1	05
	c)	In the game of baseball the objective is to win games by scoring more runs than the opposing team. Runs can only be scored if someone gets on base. Traditionally, batting average (which is actually a proportion of hits to at bats) has been used as one of the primary measures of player success. An alternative is slugging percent which is the ratio of total number of bases reached during an at bat to the number of at bats. A walk or single counts as one base, a double counts as two bases, etc. The dataset contains: • Team batting average. • Team slugging percentage, and	CO2	PO2	09

		- Team runs scored. from 10 Major League Baseball teams randomly selected from the 2012 and 2013 seasons. The followings are result gathered on the data - Mean batting average and Standard deviation for batting average. - Mean runs scored and Standard deviation for runs scored. - Making a scatter plot of team batting average and team runs scored. Identify what type of analysis (Univariate, Bivariate, and Multivariate) is performed on the data based on above information. Justify your answer.			
		OR			
5	a)	Explain the purpose of multivariate analysis, and how does it differ from univariate and bivariate analyses?	CO2	PO2	10
	b)	Explain what univariate time series data with an example is.	CO1	PO1	05
	c)	Discuss any two benefits of resampling time series data.	CO1	PO1	05
		UNIT - V			
6	a)	Explain how position (Coordinate system and Axes) on the graph can be used as a mapping aesthetic. Discuss various factors to be considered while defining data units in the positional scale on the graph.	CO1	PO1	10
	b)	Discuss commonly used aesthetics in data visualization and what type of data it can capture?	CO1	PO1	05
	c)	You are provided with a dataset containing information about the sales performance of three different products (A, B, and C) across two different regions (North and South) for the past month. The dataset has the following columns: Product: The type of product (A, B, or C). Region: The region where the sales occurred (North or South). Sales: The number of units sold. What is the appropriate visualization to represent the above dataset. Provide your justification.	CO2	PO2	05
		OR			
7	a)	Explain the purpose of using bins in a histogram when visualizing data distribution and how does the choice of bin width in a histogram impact the interpretation of the data distribution with an example?	CO2	PO2	10
	b)	Discuss Empirical Cumulative Distribution Functions and Q-Q Plots.	CO1	PO1	10

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

February / March 2024 Semester End Main Examinations

Programme: B.E.

Branch: Computer Science and Engineering

Course Code: 20CS5PCCON

Course: Computer Networks

Semester: V

Duration: 3 hrs.

Max Marks: 100

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

UNIT - I

- 1
 - a) Name the layers which implements same functionalities in OSI Model and explain the same with reasons. **4**
 - b) Explain the process of Interleaving in TDM. **4**
 - c) Draw the graph of the Manchester scheme using each of the following data streams, assuming that the last signal level has been positive. **6**
 - a. 00000000
 - b. 11111111
 - c. 01010101
 - d. 00110011
 - d) Ten sources, six with a bit rate of 200 kbps and four with a bit rate of 400 kbps are to be combined using multilevel TDM with no synchronizing bits. Answer the following questions about the final stage of the multiplexing: **6**
 - a. What is the size of a frame in bits?
 - b. What is the frame rate?
 - c. What is the duration of a frame?
 - d. What is the data rate?

UNIT - II

- 2
 - a) Illustrate the need for imposing restrictions on Ethernet frame length and Explain 802.3 MAC Frame format **10**
 - b) Design a bidirectional algorithm for the Simplest Protocol using piggybacking. Note that the both parties need to use the same algorithm. **6**
 - c) A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time(including the delays in the devices and ignoring the time needed to send a jamming signal) is 25.6 μ s, what is the minimum size of the frame? **4**

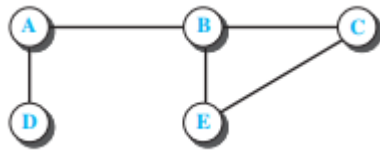
Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

OR

- 3 a) Given the dataword 1010011110 and the divisor 10111, Use polynomials and show 6
- i) The generation of the codeword at the sender site.
ii) The checking of the codeword at the receiver site.
- b) Compare byte stuffing with bit stuffing 6
- c) Explain the role of NAV CSMA/CA with a neat diagram. 8

UNIT - III

- 4 a) Consider the below graph, apply the Path vector algorithm for the same. Draw the spanning tree and Path vector table for all the nodes in the graph. 8



- b) In distance-vector routing, bad news (increase in a link metric) will propagate slowly. In other words, if a link distance increases, sometimes it takes a long time for all nodes to know the bad news. We assume that a four-node internet is stable, but suddenly the distance between nodes B and C, which is currently 2, is increased to infinity (link fails). Show how this bad news is propagated, and find the new distance vector for each node after stabilization. 8
- c) What is the need for TTL in IP header format? Illustrate with an example 4

OR

- 5 a) An organization is granted a block of addresses with the beginning address 14.24.74.0/24. The organization needs to have 3 subblocks of addresses to use in its three subnets. One subblock of 10 addresses, one subblock of 60 addresses, and one subblock of 120 addresses. Design the subblocks. 6
- b) Compare and contrast the IPv4 header with the IPv6 header. Create a table to compare each field and explain the new fields in IPV6. 8
- c) Consider sending a 3400-byte datagram into a link that has an MTU of 700 bytes. Suppose the original datagram is stamped with the identification number 520. How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation? 6

UNIT - IV

- 6 a) In a network using the Go-Back-N protocol with $m = 3$, calculate the window size. The values of variables are $S_f = 74$, $S_n = 78$, and $R_n = 76$. Assume that the network does not duplicate or reorder the packets. 6
- a. What are the sequence numbers of data packets in transit? Explain
b. What are the acknowledgment numbers of ACK packets in transit? Explain
c. Draw a diagram depicting the above scenario.

- b) Analyze the situation. Alice and Bob have no access to telephones or the Internet (think about the old days) to establish their next meeting at a place far from their homes. **8**
- i. Suppose that Alice sends a letter to Bob and defines the day and the time of their meeting. Can Alice go to the meeting place and be sure that Bob is there?
 - ii. Suppose that Bob responds to Alice's request with a letter and confirms the date and time. Can Bob go to the meeting place and be sure that Alice is there?
 - iii. Suppose that Alice responds to Bob's letter and confirms the same date and time. Can either one go to the meeting and be sure that the other person is there?
 - iv. Analyse and conclude the drawback of this situation and how it can overcome
- c) Analyse Tahoe TCP for the following scenario with a diagram: set the *ssthresh* value for 16 MSS. The station is now in the slow-start state with *cwnd* = 1 MSS and timeout occurs after third RTT. TCP sets new *ssthresh* = 4 MSS. provide the values of *cwnd*, *ssthresh*, and the current and the next state of the station after the following events: three duplicate ACKs arrives when *cwnd* = 12 MSS, followed by non duplicate ACKs, and followed by a connection termination after RTT 20 **6**

UNIT - V

- 7 a) Consider the URL for the base HTML file and 3 JPEG images residing on the same server, <http://www.bmsce.ac.in/cse/home.html>. Analyze and illustrate the steps involved in transferring a webpage from Server to the Client in case of non-persistent and persistent connections. **8**
- b) In FTP, a user (Jane) wants to retrieve an EBCDIC file named *huge* from */usr/users/report* directory using the ephemeral port 61017. The file is so large that the user wants to compress it before it is transferred. Show all the commands and responses. **6**
- c) Differentiate between Recursive Resolution and Iterative Resolution with a neat diagram **6**
